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Practical 10

Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

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Date of Performance:

Date of Performance:

CODE :

```
M ~  
Shweta Bamcode@Shweta MSYS ~  
$ nano SSTF.c
```

```
M ~  
GNU nano 8.7 SSTF.c  
#include <stdio.h>  
#include <stdlib.h>  
#include <limits.h>  
  
int main() {  
    int request[8] = {98, 183, 41, 122, 14, 124, 65, 67};  
    int visited[8] = {0};  
    int head = 53;  
    int total = 0;  
    int n = 8;  
  
    for (int i = 0; i < n; i++) {  
        int min = INT_MAX;  
        int index = -1;  
  
        for (int j = 0; j < n; j++) {  
            if (visited[j] == 0) {  
                int distance = abs(head - request[j]);  
  
                if (distance < min) {  
                    min = distance;  
                    index = j;  
                }  
            }  
        }  
  
        visited[index] = 1;  
        printf("Head moves from %d to %d with seek %d\n", head, request[index], min);  
  
        total += min;  
        head = request[index];  
    }  
  
    printf("\nTotal Head Movement = %d\n", total);  
    return 0;  
}
```

OUTPUT:

```
Shweta Bambode@Shweta MSYS ~  
$ gcc SSTF.c -o SSTF  
  
Shweta Bambode@Shweta MSYS ~  
$ ./SSTF  
Head moves from 53 to 41 with seek 12  
Head moves from 41 to 65 with seek 24  
Head moves from 65 to 67 with seek 2  
Head moves from 67 to 98 with seek 31  
Head moves from 98 to 122 with seek 24  
Head moves from 122 to 124 with seek 2  
Head moves from 124 to 183 with seek 59  
Head moves from 183 to 14 with seek 169  
  
Total Head Movement = 323
```